

BITS Pilani Digital — CodeForge V1.0

Official Bug Fix Log & Technical Enhancements Report

BITS ID: 2026eb1100208

Event: CodeForge V1.0 Debug. Reimagine. Deploy.

Application: Advanced Grading Console

Stack: HTML5, CSS3 Modern Flex/Grid, Vanilla JavaScript, SheetJS

1. Tabular Bug Fix Log

Bug #	Component	Issue Description	Root Cause Analysis	Fix Applied	Verification Result
BUG-01	Statistics Panel (Lines 220–223)	Inverted Min and Max Values: Maximum value displayed under "Min" label and minimum value displayed under "Max".	HTML element ID mapping was flipped: <code><b id="max"></code> was placed under Min, and <code><b id="min"></code> under Max.	Renamed IDs to <code>statMin</code> , <code>statMax</code> , <code>statAvg</code> , <code>statMed</code> and mapped accordingly in JavaScript.	Verified with student marks array <code>[15, 45, 92]</code> . Min displays 15, Max displays 92. Passed.
BUG-02	File Parser (Lines 312–322)	Stale & Duplicate Course Options: Courses duplicate hundreds of times on sheet re-upload.	Course dropdown was never cleared on change, and entries were added per-row without set deduplication.	Cleared <code>courseSelect.innerHTML</code> and applied <code>[...new Set(courses)]</code> deduplication.	Tested with 150-row sheet spanning 2 courses; dropdown lists exactly 2 distinct courses. Passed.
BUG-03	Canvas Bell Curve (Lines 425–439)	Divide-by-Zero Crash (NaN): If all student marks are identical, the canvas freezes completely.	When all marks are identical, standard deviation $\sigma = 0$. Gaussian formula divides by σ , generating Infinity/NaN.	Added zero-division safety guard: <code>if (std <= 0.001) return;</code>	Tested with dataset where all scores = 75. Canvas renders bars without error. Passed.

Bug #	Component	Issue Description	Root Cause Analysis	Fix Applied	Verification Result
BUG-04	Session Timer (Lines 275–294)	Initial 1-Second Timer Delay: Timer text remains static at "00:00" for the first 1000ms.	setInterval delays first invocation until after 1000ms elapses.	Extracted updateTimer() and invoked it immediately upon DOM initialization before starting interval.	Timer updates instantly upon page load without delay. Passed.
BUG-05	File Input Filter (Line 188)	Modern Excel Rejection: File picker strictly forced .xls, blocking .xlsx and .csv.	Input tag had hardcoded accept=".xls" attribute.	Updated attribute to accept=".xls,.xlsx,.csv" and implemented drag-and-drop dropzone.	Ingested modern .xlsx and .csv files successfully. Passed.
BUG-06	Reset Action (Lines 374–376)	Double Confirm Popup Spam: Clicking "Reset Range" showed two consecutive prompt dialogues.	Handler had redundant back-to-back confirm() calls.	Consolidated into a single clean confirmation dialogue.	Single confirmation prompt resets cutoffs seamlessly. Passed.
BUG-07	Data Casting (Lines 393–404)	String Concatenation Arithmetic Error: Excel numeric fields read as strings led to sorting flaws.	Missing explicit type casting on d["Total Marks"].	Wrapped values in Math.round(Number(d["Total Marks"] 0)).	Verified with string numbers like "85". Categorized properly. Passed.
BUG-08	CSV Export (Lines 472–485)	CSV Injection / Alignment Breakage: Commas in instructor or course names corrupted output columns.	Unquoted string template concatenation in CSV builder.	Added RFC 4180 compliant quotation escaping: "\${str.replace(/"/g, '""')}".	Verified with course names containing commas; opens perfectly in Excel. Passed.

2. Top 3 Meaningful Enhancements

1. Statistical Gaussian Relative Auto-Grading ($\mu \pm \sigma$)

Calculates class Mean (μ) and Standard Deviation (σ) dynamically and populates optimal relative grading cutoffs following BITS Pilani relative grading policies in 1 click ($A: \geq \mu + 1.5\sigma$, $A-: \geq \mu + 1.0\sigma$, $B: \geq \mu + 0.5\sigma$, etc.). Eliminates manual guesswork for professors.

2. Interactive Real-Time Student Inspector & Roster Preview

Embedded a live, searchable, and filterable student roster table below the workspace showing BITS ID, Total Marks, assigned Grade badge, and class percentile rank. Allows instructors to audit borderline cases instantly.

3. 1-Click "Load Sample BITS Dataset" & Universal Drag-and-Drop Dropzone

Evaluators and faculty can test the application instantly with realistic synthetic BITS course data (CS F111, EEE F111) with a single click, without needing to create or find an Excel file.